

Random and Adversarial Positional-Parameter Robustness Decouple Across Positional-Encoding Families

Doko Bandur

Faculty of Technical Sciences, University of Priština, Kosovska Mitrovica, Serbia

Abstract

Positional encodings (PEs) shape how Vision Transformers convert parameter perturbations into functional failure. We compare Learned, Sinusoidal, RoPE, and ALiBi under random and adversarial perturbations using relative RMS displacement of pre-softmax attention logits (ρ_{rel}) as a common scale. Across 72 canonical checkpoints spanning ViT-B/16 on CIFAR-100 and ImageNet-100 and ViT-S/16 on ImageNet-100, the mapping from displacement to classification damage depended on PE family and architecture. The noise-minus-adversarial nAUC gap was positive for every seed and family in every dataset-backbone cohort. A separate 18-checkpoint ViT-S/16 AMP cohort enabled an AMP-matched comparison with ViT-B/16. ViT-S/16 had lower adversarial nAUC for Learned, Sinusoidal, and RoPE but nearly unchanged noise nAUC; the architecture effect varied by family (exact Friedman $p = 0.02894$) and reversed the Learned–RoPE ordering. Within ViT-S/16, family rankings were preserved across AMP and full-FP32 training, but formal ρ_{50} equivalence held only for Sinusoidal. All six ViT-S/16 AMP ALiBi runs diverged, although the same recipe was stable for ViT-B/16 AMP and ViT-S/16 full-FP32, revealing an architecture-dependent trainability boundary. At one ViT-S/16 Sinusoidal checkpoint and budget 0.0095, direct-displacement optimization produced larger displacement ($\rho_{\text{rel}} = 0.1021$ versus 0.0938) yet retained 76.70% versus 7.36% accuracy under task-loss optimization. An all-restart audit ruled out restart selection as the cause of the Sinusoidal task-loss frontier; layerwise profiles showed objective-dependent redistribution. Together, these results show that architecture, objective, training regime, random stability, and global displacement require separate evaluation.

Keywords: Vision Transformer, positional encoding, adversarial parameter

1. Introduction

Positional information is indispensable to Vision Transformers (ViTs), whose self-attention operation is otherwise permutation equivariant with respect to the input-token sequence. Learned absolute embeddings, deterministic sinusoidal tables, rotary positional encoding (RoPE), and attention with linear biases (ALiBi) serve a common functional purpose but enter the computation graph through fundamentally different operations. Learned and sinusoidal encodings are added to token representations, RoPE changes query-key geometry through position-dependent rotations, and ALiBi contributes a position-dependent bias directly to attention logits.

This diversity makes native parameter-space robustness comparisons non-commensurate. Equal parameter norms across these mechanisms need not produce equal internal changes because the spaces differ in dimensionality, sharing structure, scale, and mapping into attention. We therefore measure the achieved relative root-mean-square displacement of pre-softmax attention logits, denoted ρ_{rel} , and analyse accuracy as a function of that achieved displacement rather than native perturbation budget alone.

A second distinction concerns perturbation direction. Random parameter noise samples typical directions from a prescribed distribution, whereas an adversarial perturbation searches for directions that increase task loss. Equal ρ_{rel} controls aggregate displacement magnitude but not its orientation, layer allocation, head or token structure, or alignment with task-critical directions. Consequently, random robustness can fail as a proxy for task-aligned robustness even after functional magnitude is matched.

We evaluate Learned, Sinusoidal, RoPE, and ALiBi in a canonical AMP-trained ViT-B/16 cohort on CIFAR-100 and ImageNet-100 and in an independent full-FP32 ViT-S/16 ImageNet-100 replication. Each positional-encoding family uses six trained seeds. A separate completed ViT-S/16 AMP cohort for Learned, Sinusoidal, and RoPE enables an AMP-matched cross-architecture comparison. Within an architecture, random and adversarial curves are compared on the same checkpoints and seed identities. Across architectures, equal seed labels provide seed-aligned descriptive contrasts but not strict training pairs because initialization dimensionality and optimization trajectories differ. We organize the three available contrasts as a factorial decomposition: the

four-family canonical cross-cohort comparison jointly changes architecture and training precision; the within-ViT-S/16 training-regime comparison holds architecture fixed and varies training precision; and the AMP-matched cross-architecture comparison holds AMP training fixed and varies architecture for the three shared families. This structure separates system-level replication, precision transfer, and backbone effects without treating equal seed labels as strict training pairs.

The ViT-B/16 results show a positive noise-minus-adversarial robustness gap in every seed, dataset, and family, together with dataset- and family-dependent gap magnitudes and a wider-range CIFAR-100 ordering reversal. The ViT-S/16 replication preserves the random-noise ranking but changes the adversarial ordering substantially. In particular, Learned PE is strongest adversarially for ViT-B/16 but falls below ALiBi and RoPE for ViT-S/16, while ALiBi changes from the weakest ViT-B/16 adversarial family to the strongest ViT-S/16 family over the locked architecture-common support.

We additionally audit the interpretation of Sinusoidal robustness. An all-restart task-loss audit records every restart rather than only the loss-selected point, and a separate direct- ρ optimization asks how far attention logits can be moved when displacement itself is optimized. These experiments show that the observed boundary is a *task-loss frontier*, not a structural upper bound: direct- ρ optimization reaches much larger displacement, while perturbations with similar global displacement can have radically different classification consequences. A layerwise decomposition shows that direct- ρ and task-loss directions allocate displacement differently across Transformer blocks.

The principal contributions are:

- an achieved attention-logit displacement scale for comparing perturbations across structurally different positional mechanisms;
- a 72-checkpoint canonical evaluation covering AMP-trained ViT-B/16 on two datasets and a full-FP32 ViT-S/16 ImageNet-100 replication, plus an 18-checkpoint ViT-S/16 AMP auxiliary cohort;
- a factorial architecture-precision decomposition showing that random-noise ranking can remain stable while adversarial ranking, mean backbone effects, and seed dispersion depend on positional-encoding family;
- an all-restart and direct-displacement analysis establishing an objective-specific Sinusoidal task-loss frontier and separating global displacement magnitude from functional damage; and

- local and layerwise analyses that distinguish parameter-to-attention amplification, task-selective harmfulness, and inter-layer displacement allocation.

2. Related Work

2.1. Positional encoding in Transformers and Vision Transformers

Self-attention does not encode token order by itself, motivating the addition of positional information to the original Transformer architecture. Vaswani et al. (2017) introduced fixed sinusoidal positional encodings and also considered learned positional embeddings. Vision Transformer subsequently adopted learned absolute positional embeddings for sequences of image patches (Dosovitskiy et al., 2021). Rotary positional encoding (RoPE) injects relative-position structure through position-dependent rotations of queries and keys (Su et al., 2024), whereas Attention with Linear Biases (ALiBi) adds head-dependent distance penalties directly to the pre-softmax attention scores (Press et al., 2022).

Vision-specific studies show that positional information can be supplied through substantially different computational mechanisms. Image relative position encoding (iRPE) systematically compared absolute and relative schemes and introduced directional two-dimensional relative encodings for vision attention (Wu et al., 2021b). Conditional positional encoding instead generates position information from the local token neighbourhood, making the encoding input dependent and improving extrapolation to unseen input sizes (Chu et al., 2023). Swin Transformer uses learned relative position biases inside shifted local windows, illustrating how positional bias can be integrated with hierarchical and locality-constrained attention (Liu et al., 2021). Conversely, CvT showed that explicit positional embeddings can become unnecessary when convolutional token embedding and projection already provide a spatial inductive bias (Wu et al., 2021a).

These works are relevant because they establish that “positional encoding” is not a single architectural operation: additive token embeddings, query–key rotations, attention-score biases, input-conditioned generators, and convolutionally supplied spatial priors have different parameterisations and different locations in the computation graph. Their main evaluations concern accuracy, resolution or length generalisation, and efficiency. The present work asks a complementary post-training question: whether structurally different PE

mechanisms preserve the same family ordering under random and task-aligned perturbations when the resulting functional displacement is matched.

2.2. Input-space adversarial robustness of Vision Transformers

Most adversarial-robustness research studies perturbations of the model input. Early gradient-based attacks demonstrated that small, norm-constrained input changes can produce large prediction changes (Goodfellow et al., 2015), and projected-gradient methods established a widely used first-order framework for evaluating and improving worst-case input robustness (Madry et al., 2018).

Several studies have examined how Vision Transformers behave under this threat model. Mahmood et al. (2021) evaluated ViTs under multiple white-box and black-box attacks and analysed transferability between convolutional and Transformer models. Naseer et al. (2021) broadened the comparison to occlusions, domain shifts, natural corruptions, spatial permutations, and adversarial perturbations, linking some observed robustness to the dynamic receptive fields of self-attention. Bai et al. (2021) showed that claims of a universal Transformer robustness advantage can disappear when model scale and training recipes are controlled, making the evaluation protocol itself part of the robustness conclusion.

Attack design can also reverse an apparent architecture ranking. Patch-Fool uses attention-aware patch optimisation and demonstrated that ViTs are not uniformly more robust than CNNs under attacks adapted to their patch and attention structure (Fu et al., 2022). Related work on adversarially training ViTs further showed that robustness depends on pretraining, optimisation choices, and how gradients or perturbations are distributed across attention blocks and image patches (Mo et al., 2022). Patch-based attacks and defences have likewise exposed vulnerabilities tied specifically to token and attention structure (Liu et al., 2023).

This literature is important to our motivation because it demonstrates that robustness rankings depend on the threat model and experimental controls, rather than following automatically from a broad architecture label. However, these studies perturb pixels or patches while leaving the trained positional mechanism unchanged. In our setting the image is fixed, all non-positional parameters are frozen, and the attack surface is the PE-specific parameter or inference-time positional state.

2.3. Robustness to neural-network parameter perturbations

Perturbations of model parameters have been studied from optimisation, generalisation, and deployment-security perspectives. Adversarial Weight Perturbation (AWP) regularises adversarial training by optimising against locally harmful weight changes and relates the weight-loss landscape to robust generalisation (Wu et al., 2020). Theoretical work has analysed how bounded weight perturbations affect margins, generalisation, and adversarial robustness (Tsai et al., 2021). Random and multiplicative weight perturbations have also been used as training-time regularisers for robustness to common corruptions (Trinh et al., 2024).

The distinction between random and objective-aligned weight changes has also appeared in optimisation research. Random weight perturbation optimises an expected neighbourhood objective, whereas SAM/AWP-style methods emphasise a locally worst-case neighbourhood; Li et al. (2024) directly compared these training-time paradigms and showed that their behaviour depends strongly on perturbation magnitude and optimisation design. This is conceptually relevant to our random-versus-adversarial comparison, but their objective is to improve training and generalisation, not to evaluate the post-training robustness ordering of heterogeneous PE mechanisms.

From a security perspective, highly targeted post-training changes can be far more damaging than unstructured noise. The Bit-Flip Attack identified a small number of vulnerable stored weight bits that can collapse network accuracy, while substantially more random flips have little effect (Rakin et al., 2019). Small parameter changes can also implant backdoors while preserving clean behaviour (Garg et al., 2020), and adversarial parameter attacks can selectively degrade a deployed model’s robustness without an equally conspicuous loss of clean accuracy (Yu et al., 2023). These results provide an important qualitative precedent for our central question: stability under random parameter noise need not characterise sensitivity to task-aligned directions.

Most prior parameter-space methods nevertheless perturb broad collections of weights or use perturbations as a training mechanism. Their magnitudes are typically specified in the native parameter space. That convention is appropriate within one parameterisation, but it does not directly support comparisons between Learned, Sinusoidal, RoPE, and ALiBi, whose positional states have different dimensions, constraints, sharing patterns, and direct effects on attention. We therefore freeze all non-positional parameters and

compare perturbations through the attention-logit displacement they actually produce.

2.4. *Random and adversarial directions in local model geometry*

Loss-landscape research provides useful context for interpreting random and objective-aligned parameter directions. Large-batch training was associated empirically with sharper minima and weaker generalisation (Keskar et al., 2017), while Li et al. (2018) developed filter-normalised random-direction visualisations to make landscape comparisons less sensitive to layer scale. Sharpness-Aware Minimisation explicitly optimises a worst-case loss in a parameter neighbourhood (Foret et al., 2021), illustrating why an objective-aligned direction and a random direction probe different properties even at nominally similar parameter magnitudes.

At the same time, parameter-space sharpness is not invariant to benign reparameterisations. Dinh et al. (2017) constructed functionally equivalent networks with arbitrarily different apparent sharpness, showing why native-coordinate geometry must be interpreted with care. This observation is especially relevant when comparing PE families whose parameters have different semantics: equality of a raw RMS or ℓ_p perturbation does not imply equality of the induced change in the network computation.

Our study does not estimate a full Hessian, a full Jacobian singular-value decomposition, or a global worst case. Instead, isotropic Gaussian perturbations sample typical directions in each PE-specific parameter space, while task gradients and restarted PGD search objective-aligned harmful directions. We compare the two regimes on a shared functional metric—achieved pre-softmax attention-logit displacement—and treat the independently trained checkpoint/seed as the unit of replication. The local geometry probe then separates parameter-to-attention amplification from classification damage per unit of actually achieved displacement.

2.5. *Position of the present study*

The closest literature can therefore be distinguished along three axes. First, the *threat model* differs: standard ViT robustness studies modify pixels or patches, whereas we modify only the positional mechanism of an otherwise frozen trained model. Second, the *comparison metric* differs: parameter-attack and landscape studies usually constrain a native weight norm, whereas we match the achieved displacement in pre-softmax attention logits across heterogeneous PE parameterisations. Third, the *statistical unit* differs: random

directions are repeated measurements within a checkpoint, not independent replicates; inference is performed on six shared training-seed identities per family. This combination enables a direct test of whether random positional-parameter stability predicts adversarial positional robustness, rather than merely confirming that a gradient-aligned perturbation is stronger than a random one.

3. Functional Displacement Framework

3.1. Notation

Let

$$f(x; \theta, p)$$

denote a Vision Transformer with ordinary model parameters θ and positional parameters or positional state p . The meaning of p is PE-family dependent: it denotes an additive embedding tensor for Learned and Sinusoidal PE, a rotation phase representation for RoPE, and the shared head-specific slopes used to generate attention biases for ALiBi.

For input x , let

$$Z_\ell(x; \theta, p) \in \mathbb{R}^{H \times N \times N}$$

denote the pre-softmax attention logits in Transformer block ℓ , where H is the number of heads and N is the token count. Given a perturbation δ of the positional state, define

$$\Delta Z_\ell(x; \delta) = Z_\ell(x; \theta, p + \delta) - Z_\ell(x; \theta, p).$$

For RoPE, the notation $p + \delta$ represents an additive phase perturbation followed by reconstruction of the corresponding sine and cosine tensors. This preserves the unit-circle constraint of every rotation pair. For ALiBi, the perturbed slopes are used to regenerate the complete position-dependent attention-bias tensor.

3.2. Family-specific perturbation spaces

The native perturbation variable contains only the unique positional state coordinates. Learned and Sinusoidal PE use a single additive perturbation $\delta \in \mathbb{R}^{N \times d}$ at the Transformer input. Although every Transformer block instantiates its own RoPE phase cache and applies its own copy of the same ALiBi slope vector, the analysis treats these blockwise instances as tied copies

of one shared positional state. A single RoPE phase perturbation $\delta \in \mathbb{R}^{N \times d_h/2}$ and a single ALiBi slope perturbation $\delta \in \mathbb{R}^H$ are applied identically in all $L = 12$ blocks. The global RMS constraint is computed over these unique perturbation coordinates rather than over their repeated blockwise instances.

Perturbations act directly on the cached RoPE phase tensor. The underlying inverse-frequency buffer that generates the clean phases is held fixed and is excluded from the perturbation space. Thus, the RoPE threat model permits position-specific phase corruption but does not perturb the shared frequency law that couples clean phases across token positions. For ALiBi, the precomputed distance matrix is likewise fixed; only the shared slope vector is perturbed.

The resulting numbers of scalar perturbation degrees of freedom are Nd for Learned and Sinusoidal PE, $Nd_h/2$ for RoPE, and H for ALiBi. They are, respectively, 151,296, 151,296, 6,304, and 12 on ImageNet-100 ($N = 197$), and 49,920, 49,920, 2,080, and 12 on CIFAR-100 ($N = 65$).

3.3. Achieved attention-logit displacement

We define the absolute attention-logit displacement as

$$\rho_{\text{abs}}(\delta) = \left[\frac{1}{L} \sum_{\ell=1}^L \frac{1}{|\mathcal{C}|HN^2} \sum_{x \in \mathcal{C}} \|\Delta Z_{\ell}(x; \delta)\|_F^2 \right]^{1/2}, \quad (1)$$

where $\mathcal{C}$ is the fixed calibration set and each Transformer block receives equal weight.

Because the overall scale of clean attention logits differs across models and PE families, we normalize Eq. (1) by the clean-logit RMS:

$$s_Z = \left[\frac{1}{L} \sum_{\ell=1}^L \frac{1}{|\mathcal{C}|HN^2} \sum_{x \in \mathcal{C}} \|Z_{\ell}(x; \theta, p)\|_F^2 \right]^{1/2}. \quad (2)$$

The relative displacement is

$$\rho_{\text{rel}}(\delta) = \frac{\rho_{\text{abs}}(\delta)}{s_Z}. \quad (3)$$

The quantity ρ_{rel} is dimensionless and controls the aggregate magnitude of the induced change in pre-softmax attention logits. It does not constrain the direction of that change or its distribution across layers, heads, queries, and keys.

3.4. Random and adversarial perturbation regimes

For the random regime, we sample Gaussian directions in the unique PE parameter space and normalize them using the global root-mean-square norm. A fixed sampled direction is retained across the parameter-budget grid, producing a coherent per-draw path through parameter space. Random curves are first averaged within each trained seed before any across-seed summary is computed.

For the adversarial regime, we solve

$$\max_{\delta} \frac{1}{|\mathcal{A}|} \sum_{(x,y) \in \mathcal{A}} \mathcal{L}(f(x; \theta, p + \delta), y) \quad \text{subject to} \quad \|\delta\|_{\text{RMS}} \leq \varepsilon, \quad (4)$$

where $\mathcal{A}$ is disjoint from the calibration and evaluation sets. The optimization is performed using projected gradient ascent with multiple random restarts.

Equation (4) is constrained in the native PE parameter space, not directly in ρ -space. Every resulting perturbation is therefore reparameterised after optimization by its achieved value of ρ_{rel} from Eq. (3).

3.5. Robustness curves and normalized AUC

For each trained model and perturbation regime, let

$$a(\rho) = \frac{\text{Acc}(\rho)}{\text{Acc}_{\text{clean}}}$$

denote normalized classification accuracy. Normalized accuracy is not clipped to the interval $[0, 1]$. Values slightly above one can occur when a perturbed model marginally outperforms its own clean reference on the finite evaluation subset; these fluctuations are retained symmetrically rather than truncated. We include the exact clean anchor

$$(\rho, a) = (0, 1).$$

Because PGD directly maximizes classification loss and finite optimization may yield small non-monotonic fluctuations across native budgets, we construct the adversarial curve using the monotone lower envelope in increasing achieved ρ . This represents the strongest observed attack up to each displacement level.

For a common interval $[0, \rho_{\text{max}}]$, the normalized area under the robustness curve is

$$\text{nAUC} = \frac{1}{\rho_{\text{max}}} \int_0^{\rho_{\text{max}}} a(\rho) d\rho. \quad (5)$$

The normalization by $\rho_{\max}$ makes the statistic interpretable on the same $[0, 1]$ scale as normalized accuracy. Integration is performed separately for every trained seed; means, uncertainty estimates, and paired regime differences are computed only after the per-seed nAUC values have been obtained.

The primary analysis uses $\rho_{\max} = 0.09$, with $\rho_{\max} = 0.095$ reported as an interval-sensitivity analysis. The primary and sensitivity endpoints are defined from the common achieved-displacement support of ImageNet-100, the limiting dataset, and are applied to both datasets to preserve identical integration intervals. The separate wider-range CIFAR-100 endpoint used in Section 5 is defined from the CIFAR-100 common support. The endpoint-selection criteria are described in Section 4.8.

4. Experimental Setup

4.1. Datasets and evaluation splits

We conduct experiments on CIFAR-100 and ImageNet-100. For every dataset, split seed 0 is used to construct a deterministic permutation and mutually disjoint calibration, attack, and evaluation subsets.

For ImageNet-100, the validation set contains 5000 images. We use 256 images for attention-logit displacement calibration, 1280 images for adversarial parameter optimization, and the remaining 3464 images for accuracy evaluation.

For CIFAR-100, the test set contains 10000 images. We use 256 images for attention-logit displacement calibration, 1280 images for attack optimization, and the remaining 8464 images for accuracy evaluation.

The calibration, attack, and evaluation subsets have zero pairwise overlap.

4.2. Positional-encoding families

Let P denote the number of image patches and $N = P + 1$ the total sequence length including the [CLS] token. The convolutional patch map is flattened in row-major raster order and the classification token is then prepended. Thus, sequence position $p = 0$ denotes [CLS], whereas positions $p = 1, \dots, P$ denote image patches in row-major order. None of the four baseline implementations uses an axial, factorized, or otherwise explicitly two-dimensional positional construction; image geometry enters these mechanisms through the one-dimensional raster sequence.

We study four positional-encoding mechanisms:

- **Learned:** a trainable absolute positional table $E_{\text{pos}} \in \mathbb{R}^{1 \times N \times d}$ is added to the token representations. Its entries, including the independent [CLS] entry at $p = 0$, are initialized from $\mathcal{N}(0, 0.02^2)$;
- **Sinusoidal:** a deterministic one-dimensional sinusoidal table is added to the token representations,

$$E_{\text{pos}}(p, 2k) = \sin(p \cdot 10000^{-2k/d}), \quad (6)$$

$$E_{\text{pos}}(p, 2k + 1) = \cos(p \cdot 10000^{-2k/d}). \quad (7)$$

Consequently, [CLS] receives the $p = 0$ vector $(0, 1, 0, 1, \dots)$;

- **RoPE:** rotary positional encoding uses the same one-dimensional raster index. With head dimension $d_h = 64$, the clean phase is $\theta_{p,k} = p \cdot 10000^{-2k/d_h}$ for $k = 0, \dots, 31$. Each block stores sine and cosine caches of shape $1 \times 1 \times N \times 32$, broadcast across all heads. Rotation uses the half-split convention, pairing the first and second halves of each head vector. The $p = 0$ [CLS] rotation is therefore the identity;
- **ALiBi-style:** a bidirectional, symmetric linear bias is added to the attention logits,

$$B_{ij}^{(h)} = -m_h |i - j|, \quad m_h = 2^{-8(h+1)/H}, \quad h = 0, \dots, H - 1, \quad (8)$$

where $H = 12$ for ViT-B/16 and $H = 6$ for ViT-S/16. The same architecture-specific fixed slope vector is used in every block and distance is computed over the complete one-dimensional sequence. No special [CLS] rule is applied, so its distance to patch position j is j .

Equation (8) is a monotone geometric slope schedule, not the reference non-power-of-two `get_slopes(12)` construction. For brevity, the tables and figures use the label “ALiBi” to denote this precisely defined ALiBi-style implementation. For the strongest-slope head, $m_0 \approx 0.630$ in ViT-B/16 and $m_0 \approx 0.397$ in ViT-S/16; on ImageNet-100 the corresponding [CLS]-to-final-position biases are approximately -123.5 and -77.8 . These values impose architecture-specific head priors toward early indexed raster positions, although the realized attention also depends on the content-dependent query-key logits.

4.3. Architecture and training

The canonical ViT-B/16 backbone uses embedding dimension $d = 768$, 12 Transformer blocks, 12 attention heads, head dimension 64, an MLP expansion ratio of 4, and dropout probability 0.1. The independent ViT-S/16 ImageNet-100 replication uses $d = 384$, 12 blocks, 6 heads, the same head dimension 64, MLP ratio 4, and dropout probability 0.1. Within each architecture, models differ only in their positional-encoding mechanism.

For ImageNet-100, images are evaluated at 224×224 with 16×16 patches, producing 196 patch tokens and sequence length $N = 197$ after adding [CLS]. CIFAR-100 is evaluated only in the ViT-B/16 cohort at native 32×32 resolution with 4×4 patches, producing 64 patch tokens and $N = 65$.

All 72 canonical checkpoints are trained from scratch for 300 epochs using AdamW with initial learning rate 3×10^{-4} , weight decay 0.1, cosine annealing, 20 warmup epochs, batch size 128, label smoothing 0.1, and Mixup with $\alpha = 0.8$. The ViT-B/16 checkpoints are trained with CUDA automatic mixed precision and dynamic loss scaling; the canonical ViT-S/16 checkpoints are trained in full FP32. An auxiliary 18-checkpoint ViT-S/16 AMP cohort (three families by six seeds) follows the same training recipe and is used only in the within-ViT-S/16 training-regime and AMP-matched cross-architecture comparisons. The optimizer contains all trainable model parameters in one group, without positional-parameter, bias, or normalization exclusions from weight decay. Thus, the Learned positional table and trainable [CLS] token receive weight decay, whereas the Sinusoidal table, RoPE phase construction, and ALiBi slope state are fixed buffers during training. All canonical and auxiliary robustness attacks and evaluations are conducted in FP32 under the locked numerical protocol; attack/evaluation precision is therefore distinct from training precision.

4.4. Checkpoint cohort and clean performance

For every positional-encoding family in each architecture–dataset cohort, we analyse six independently trained models with seeds

$$\{42, 123, 456, 789, 1011, 1213\}.$$

The study therefore contains 48 ViT-B/16 checkpoints (4 families $\times$ 6 seeds $\times$ 2 datasets) and 24 ViT-S/16 ImageNet-100 checkpoints (4×6), for a total of 72.

Table 1: Clean classification accuracy. Values are mean $\pm$ sample standard deviation over six independently trained seeds. ViT-B/16 values use the complete dataset evaluation set; ViT-S/16 values use its fixed 3464-image ImageNet-100 evaluation split.

PE family	ViT-B ImageNet-100	ViT-B CIFAR-100	ViT-S ImageNet-100
Learned	79.46 ± 0.51	67.92 ± 0.46	77.95 ± 0.41
Sinusoidal	81.62 ± 0.41	67.22 ± 0.47	78.86 ± 0.38
RoPE	84.56 ± 0.30	73.18 ± 0.19	82.64 ± 0.34
ALiBi	81.01 ± 0.36	67.76 ± 0.30	79.21 ± 0.28

Within each architecture, comparisons between perturbation regimes and families retain the same seed identities and checkpoints. Across architectures, the seed labels are aligned for descriptive contrasts, but are not treated as strict pairs because model dimensionality, initialization, and training trajectories differ. Ordinary model weights remain frozen throughout robustness evaluation; only the family-specific positional representation is perturbed.

Table 1 reports clean accuracy. ViT-B/16 values are measured on the complete ImageNet-100 validation and CIFAR-100 test sets. ViT-S/16 values are measured on the fixed 3464-image evaluation split used by its canonical robustness protocol. Robustness curves use each seed’s clean accuracy on the corresponding evaluation split as the normalization anchor.

The Learned table is the only positional state directly subject to AdamW weight decay. This optimizer asymmetry is part of the complete as-trained implementation and was not isolated as a separate causal factor.

4.5. Random parameter-noise protocol

Random robustness is evaluated using Gaussian directions sampled in the unique PE parameter space. Each direction is normalized according to the global RMS convention.

For a given model and random draw, the same direction is retained across the complete native budget grid. This produces a coherent path through parameter space rather than a collection of independently resampled perturbations.

Ten random directions are evaluated per model in the main robustness experiment. For PE family f , training seed s , native noise budget b , and fixed random directions $d = 1, \dots, 10$, the achieved displacement and normalized

accuracy are first averaged at the same native budget:

$$\bar{\rho}_{f_{sb}} = \frac{1}{10} \sum_{d=1}^{10} \rho_{f_{sbd}}, \quad \bar{a}_{f_{sb}} = \frac{1}{10} \sum_{d=1}^{10} a_{f_{sbd}}. \quad (9)$$

The budget-paired points $(\bar{\rho}_{f_{sb}}, \bar{a}_{f_{sb}})$ define one random-noise curve per trained seed. We do not integrate the ten draw-specific curves separately and do not construct a union grid over their distinct achieved-displacement coordinates.

4.6. Adversarial parameter perturbations

Adversarial PE perturbations are generated using projected gradient ascent on cross-entropy loss over the fixed attack subset. We use five independent random restarts and retain the restart with the largest attack loss.

The step size is set to a fixed fraction of the native RMS budget:

$$\alpha = 0.05\epsilon.$$

Family-specific optimization lengths were first selected in a seed-42 development audit and then subjected to a prespecified harmonization check on confirmation seed 123. For matched N -to- $2N$ comparisons, a retained schedule had to satisfy absolute changes of at most 0.003 in normalized accuracy, 0.005 in selected attack loss, and 0.003 in achieved ρ_{rel} , with identical five-restart seeds. Learned 400×5 , Sinusoidal 400×5 , and RoPE 200×5 passed at all three tested transition budgets. ALiBi 100×5 failed the accuracy tolerance at native budget 0.02; a targeted 200×5 -to- 400×5 comparison at that budget then passed all criteria. The complete matched comparisons, including the initial development sequence, are retained in the reproducibility package.

For CIFAR-100, the final attack configurations are:

- Learned: 400 steps and 5 restarts;
- Sinusoidal: 200 steps and 5 restarts;
- RoPE: 100 steps and 5 restarts;
- ALiBi: 100 steps and 5 restarts.

For ViT-B/16 ImageNet-100, the final configurations are:

- Learned: 400 steps and 5 restarts;

- Sinusoidal: 400 steps and 5 restarts;
- RoPE: 200 steps and 5 restarts;
- ALiBi: 200 steps and 5 restarts.

All six ViT-B/16 ImageNet-100 ALiBi attack curves were computed under this final 200×5 schedule.

For ViT-S/16 ImageNet-100, an architecture-specific convergence and confirmation workflow locked:

- Learned: 1600 steps and 5 restarts;
- Sinusoidal: 200 steps and 5 restarts;
- RoPE: 200 steps and 5 restarts;
- ALiBi: 200 steps and 5 restarts.

The 1600×5 Learned schedule is supported by targeted matched longer-horizon checks; no claim is made that 3200×5 or 6400×5 was executed.

The attack is constrained in the family-specific native PE parameter space using the global RMS norm. It is not directly constrained by ρ_{rel} .

4.7. Construction of the final displacement grid

The mapping from native parameter budget to achieved ρ_{rel} is family and seed dependent. A uniform native budget schedule can therefore leave long interpolation intervals in functional space, particularly close to the clean operating point. We construct the final family-specific measurement grids to provide denser coverage in these regions while preserving the same attack objective, parameter norm, optimization length, and number of restarts.

For Learned PE on both datasets, the final grid includes the targeted low-budget native RMS values

$$\{0, 0.00025, 0.0005, 0.00075, 0.001\}.$$

For CIFAR-100 RoPE, it includes

$$\{0, 0.005, 0.010, 0.015, 0.020\},$$

and for ImageNet-100 RoPE it includes

$$\{0, 0.0125, 0.0375\},$$

with the final 200-step, five-restart attack configuration. In addition, the CIFAR-100 transition-region grid contains native RMS budgets 0.00625 for Learned, 0.01125 for Sinusoidal, 0.225 for RoPE, and 0.0375 and 0.060 for ALiBi. These five budgets were selected from the seed-42 development scan before the final cohort was analysed and were subsequently measured for all six seeds using the locked family-specific schedules. A same-protocol H200 rerun of the seed-42 transition points agreed with the earlier RTX Blackwell Server Edition run to within 4.91×10^{-4} in normalized accuracy and 2.56×10^{-5} in achieved ρ_{rel} ; the homogeneous H200 values are used in the final six-seed transition grid.

The locked ViT-S/16 ImageNet-100 adversarial native-budget grids are

Learned: $\{0, 0.003, 0.004, 0.0045, 0.005, 0.006, 0.008\}$,

Sinusoidal: $\{0, 0.005, 0.006, 0.007, 0.0075, 0.008, 0.0085, 0.009, 0.0095, 0.010\}$,

RoPE: $\{0, 0.010, 0.025, 0.050, 0.075, 0.100\}$,

ALiBi: $\{0, 0.001, 0.0025, 0.005, 0.010, 0.015, 0.025, 0.040\}$.

Random-noise curves use ten fixed Gaussian directions per checkpoint and family, retained across each native-budget path. Adaptive threshold refinement, when present, changes only the nominal budget grid and does not change steps, restarts, objective, split, or checkpoint selection.

All validated intermediate points are merged with the remaining canonical measurements before curve construction. A single canonical analysis script then generates the seed-level curves, the primary and wider-range nAUC tables, and the main robustness figure from this final measurement grid. The script records SHA-256 hashes for every consumed result file, and the same curve objects feed all three manuscript outputs, preventing table-figure convention drift.

4.8. Statistical analysis

The trained checkpoint is the unit of replication. All curve construction, numerical integration, and regime comparisons are therefore performed separately for each seed before any across-seed summary is calculated. The ten random-noise draws are repeated measurements within a checkpoint and are aggregated at matched native budgets according to Eq. (9); they are not treated as independent experimental replicates.

Seed 42 served as the protocol-development checkpoint for selecting the family-specific PGD schedules and native budget grids and is therefore not

fully independent of those design choices. It remains in the prespecified six-seed cohort, but we additionally repeat the primary analysis after excluding it. In that five-seed sensitivity analysis, all 40 remaining noise-minus-adversarial gaps are positive, the family-mean orderings are preserved on both datasets, and the exact Friedman tests give $Q = 13.56$ and $p = 1.38648 \times 10^{-4}$ for each dataset (1104 of $24^5 = 7,962,624$ within-block permutations). The development audit is used only to validate the optimization protocol and is not treated as independent confirmatory evidence.

For every dataset, PE family, seed, and perturbation regime, measured points are ordered by achieved ρ_{rel} . The exact clean anchor $(0, 1)$ is then included. For adversarial perturbations, we use the cumulative lower envelope of normalized accuracy:

$$\tilde{a}_{\text{adv}}(\rho_i) = \min_{j \leq i} a_{\text{adv}}(\rho_j). \quad (10)$$

This prevents small finite-optimization or evaluation fluctuations from making a weaker attack at a larger native budget improve the strongest-observed adversarial robustness curve.

Because the envelope is applied only to the adversarial regime, it is a one-sided transformation: it can only lower the adversarial curve and can therefore only enlarge the noise-minus-adversarial gap. This aggregation asymmetry is intentional because the two regimes target different estimands. Random-noise curves estimate the mean response to isotropically sampled directions and are therefore averaged across draws at matched native budgets. Adversarial curves estimate the strongest observed optimized perturbation: the highest-loss restart is retained at each native budget, followed by the cumulative lower envelope over achieved displacement. Applying the same monotonic transformation to the noise regime would be inappropriate because random directions are not optimized and their non-monotonicities represent sampling variability rather than attack-optimization artefacts.

A deterministic sensitivity analysis without the adversarial envelope is reported in the supplementary material. After adding the exact clean anchor, the raw measured normalized accuracies are ordered by increasing achieved displacement and connected by piecewise-linear interpolation, without clipping values above one. All 48 seed-level gaps remain positive, all within-seed family rank configurations are preserved, and the largest change in any family-mean gap is small (CIFAR-100 Learned, $0.015369 \rightarrow 0.014591$).

The primary robustness statistic is the normalized area under the accuracy–

displacement curve:

$$\text{nAUC}_{s,f,r} = \frac{1}{\rho_{\max}} \int_0^{\rho_{\max}} a_{s,f,r}(\rho) d\rho, \quad (11)$$

where s denotes the training seed, f the PE family, and r the perturbation regime. Piecewise-linear interpolation and trapezoidal integration are used between adjacent achieved-displacement points. Integration endpoints were selected from the common achieved-displacement support of the measured curves, that is, from displacement intervals covered by every included seed-family curve under both perturbation regimes. Endpoint selection therefore depends only on achieved-displacement coverage and not on accuracy outcomes, family rankings, or regime differences. The locked ImageNet-100 session protocol initially specified a planned primary interval $\rho_{\text{rel}} \in [0, 0.1]$. During final analysis, the endpoint was reduced on coverage grounds: the smallest maximum achieved adversarial displacement is 0.095748 for ImageNet-100 Sinusoidal PE, while the smallest maximum achieved displacement among all noise curves in the final merged grid is 0.115553. We therefore use $\rho_{\max} = 0.09$ as a conservative primary endpoint and $\rho_{\max} = 0.095$ as a near-maximal common-support sensitivity endpoint. This is a support-driven deviation from the initially planned endpoint, not an outcome-driven change. The same two endpoints are applied to CIFAR-100 to preserve identical integration intervals across datasets. Every included curve contains at least one measured point beyond each applicable endpoint; consequently, no reported nAUC requires extrapolation beyond measured support.

For each dataset and PE family, the within-seed decoupling statistic is

$$G_{s,f} = \text{nAUC}_{s,f,\text{noise}} - \text{nAUC}_{s,f,\text{adv}}. \quad (12)$$

We report the sample mean and sample standard deviation across the six seeds. Uncertainty for mean paired differences is additionally summarized using a paired bootstrap confidence interval obtained by resampling the six seed identities with replacement and retaining the pairing between regimes.

Because only six paired observations are available, exact non-parametric inference has coarse resolution. For each dataset-family combination, we compute an exact two-sided sign-flip test under the sign-symmetry null for the six paired seed-level gaps. Conditional on the observed gap magnitudes, all $2^6 = 64$ sign assignments are enumerated. When all six differences have the same positive sign, the two-sided exact probability is

$$p = \frac{2}{2^6} = 0.03125.$$

This is the smallest attainable two-sided value in the $n = 6$ cohort. The eight family-specific tests are therefore used only as secondary checks of directional consistency, not as separate confirmatory significance tests. In particular, no individual test can meet the Bonferroni family-wise threshold $0.05/8 = 0.00625$; the corresponding Holm–Bonferroni-adjusted values are 0.25. We make no multiplicity-adjusted significance claim from these tests. Formal omnibus inference is instead provided by the dataset-level exact Friedman permutation tests described below, while effect magnitudes, pointwise paired bootstrap intervals, and positive-seed counts characterize the direction and consistency of the family-specific regime differences.

To test whether the magnitude of the noise-minus-adversarial gap differs across PE families, we apply a Friedman repeated-measures test to the four family-specific gap values within each dataset, with the six shared training-seed identities treated as repeated-measures blocks. The test therefore ranks the four family-specific gaps within each shared seed identity and evaluates whether those within-seed rankings are systematically different across families. Under the null hypothesis that the four family labels are exchangeable within each shared seed block, and because the seed-level gap values contain no ties, we compute the exact null distribution of the Friedman statistic over all $(4!)^6 = 191,102,976$ possible within-block rank assignments. We report the Friedman statistic Q and its exact permutation p -value. The three-degree-of-freedom asymptotic χ^2 approximation is retained only for reference. Family-specific paired effects and their uncertainty remain the primary quantities; the omnibus test assesses between-family heterogeneity of gap magnitude and does not test whether the gap is positive.

Because CIFAR-100 provides broader common achieved-displacement support, we analyse the interval $\rho_{\max} = 0.23$ separately as an ordering-sensitivity analysis. The endpoint is a conservative choice within the CIFAR-100 common support; the smallest per-seed maximum achieved adversarial displacement is 0.236431. This wider interval was not part of the documented primary endpoint plan and is not treated as an additional confirmatory test. It is not pooled with the primary $[0, 0.09]$ inference and is used only to assess whether conclusions about relative PE robustness persist beyond the small-displacement regime. Every included CIFAR-100 curve has measured support beyond 0.23, so no extrapolation is required.

No individual random draw, perturbation budget, attention layer, or image is treated as an independent replicate. Unless explicitly stated otherwise, reported standard deviations use Bessel’s correction (`ddof=1`).

For the four-family canonical cross-cohort ImageNet-100 comparison, support is the intersection over every architecture, family, seed, and perturbation regime included in that comparison, $\rho_{\text{canonical}} = 0.0792383973$. Because ViT-B/16 is AMP-trained and the canonical ViT-S/16 cohort is full-FP32-trained, this comparison does not isolate architecture from training precision. A separate three-family AMP-matched cross-architecture analysis therefore defines $\rho_{\text{cross}} = 0.0873455716$. Each analysis uses 0.9 and 0.75 of its own locked endpoint for sensitivity; no support is mixed and no curve is extrapolated. Near-duplicate achieved-displacement points are grouped when

$$|\rho_i - \rho_j| \leq \max(5 \times 10^{-7}, 5 \times 10^{-6} \max(|\rho_i|, |\rho_j|, 10^{-12})) .$$

Within a group, the retained canonical point is selected by lower normalized accuracy, then larger task loss, lower nominal budget, and stable source order. These rules are fixed before architecture-level summaries are computed.

Seed labels are aligned across architectures, but the two backbones consume different amounts of the global random stream during initialization. The training DataLoader has no separately locked generator, so models sharing a seed label share neither initial weights nor guaranteed data ordering. Seeds are therefore aligned labels, not matched pairs. Seed-aligned difference intervals are reported for consistency with the within-architecture analyses, and unpaired Welch intervals are reported as a supplementary sensitivity check.

Normalized accuracy is not clipped at one: a noisy model can marginally outperform its clean reference on the fixed evaluation split, so a noise-nAUC value may also lie slightly above one.

The all-restart Sinusoidal task-loss audit and the direct-displacement optimization audit are objective-specific structural analyses. They do not extend canonical ρ_{common} , ρ_{50} , nAUC, or task-loss-frontier support.

5. ViT-B/16 Random–Adversarial Robustness Decoupling

5.1. Primary robustness curves

Figure 1 shows the final merged robustness curves on the common achieved-displacement scale. The random-noise and adversarial responses separate to markedly different degrees across PE families and datasets, while the adversarial curves generally decline more rapidly over the primary range. The corresponding seed-aggregated nAUC values and paired gaps are summarized in Table 2.

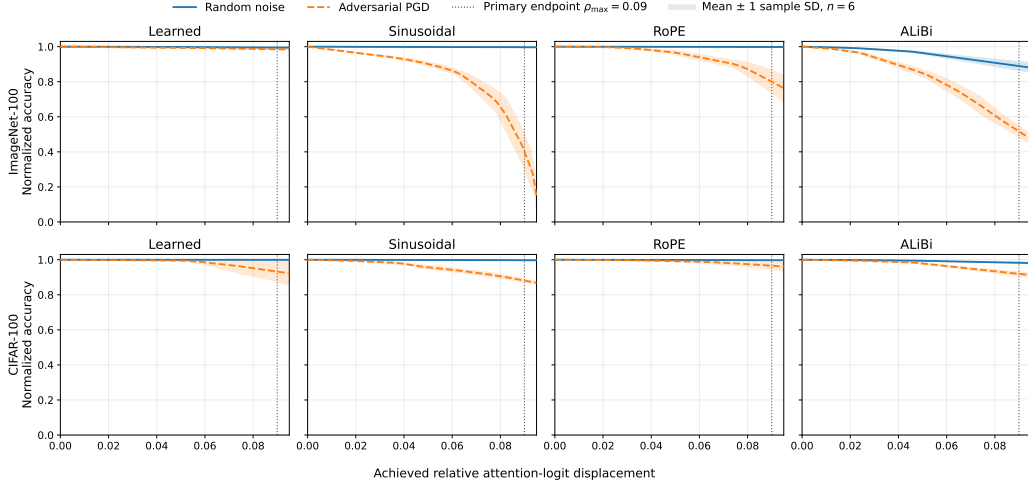


Figure 1: Normalized accuracy as a function of achieved relative attention-logit displacement for random parameter noise and adversarial positional-parameter perturbations. Solid and dashed curves show the across-seed mean, and shaded bands show mean $\pm$ one sample standard deviation over $n = 6$ independently trained seeds per PE family. The dotted vertical line marks the primary integration endpoint $\rho_{\max} = 0.09$. Curves are constructed and displayed over achieved attention-logit displacement rather than nominal native parameter budget. The random curves use the budget-paired ten-draw aggregation in Eq. (9); the figure and both nAUC tables are generated by the same canonical analysis script.

5.2. Normalized area under the robustness curve

5.3. Family-dependent robustness gaps

Across both datasets, random-noise nAUC exceeds adversarial nAUC for every PE family. This direction is consistent across all six paired seeds in each dataset–family combination.

The result is not scientifically interesting merely because an optimized attack is stronger than random perturbation. The relevant finding is that the size of the random–adversarial gap is strongly family dependent and changes across datasets.

On ImageNet-100, Sinusoidal and ALiBi exhibit much larger separations between random and adversarial robustness than Learned PE. RoPE occupies an intermediate position. On CIFAR-100, all four gaps are smaller within the primary interval, and their relative ordering differs from that observed on ImageNet-100.

Table 2: Primary normalized area under the robustness curve over $\rho_{\text{rel}} \in [0, 0.09]$. Values are mean $\pm$ sample SD across six paired training seeds. Random-noise curves use the budget-paired ten-draw aggregation defined in Section 4.8. Confidence intervals are paired seed-level bootstrap intervals for the noise-minus-adversarial gap and are pointwise, not simultaneous. The eight exact two-sided sign-flip tests each yielded the minimum attainable unadjusted value $p = 0.03125$ at $n = 6$; they are secondary directional-consistency checks and do not support multiplicity-adjusted individual significance claims (see Section 4.8).

Dataset	PE family	Noise nAUC	Adversarial nAUC	Gap	95% CI	Seeds with gap > 0
ImageNet-100	Learned	0.9980 ± 0.0011	0.9935 ± 0.0025	0.0045 ± 0.0027	[0.0027, 0.0066]	6/6
	Sinusoidal	0.9983 ± 0.0012	0.8610 ± 0.0205	0.1373 ± 0.0211	[0.1219, 0.1529]	6/6
	RoPE	0.9990 ± 0.0005	0.9516 ± 0.0134	0.0474 ± 0.0137	[0.0382, 0.0582]	6/6
	ALiBi	0.9602 ± 0.0081	0.8336 ± 0.0206	0.1266 ± 0.0231	[0.1074, 0.1390]	6/6
CIFAR-100	Learned	0.9998 ± 0.0009	0.9844 ± 0.0125	0.0154 ± 0.0122	[0.0072, 0.0244]	6/6
	Sinusoidal	0.9989 ± 0.0010	0.9584 ± 0.0043	0.0405 ± 0.0044	[0.0379, 0.0441]	6/6
	RoPE	0.9988 ± 0.0005	0.9902 ± 0.0060	0.0087 ± 0.0060	[0.0044, 0.0131]	6/6
	ALiBi	0.9935 ± 0.0010	0.9733 ± 0.0046	0.0203 ± 0.0037	[0.0175, 0.0229]	6/6

The interaction between PE family and perturbation regime therefore indicates that random parameter-noise stability is not a uniform surrogate for adversarial positional-parameter robustness.

The between-family heterogeneity of gap magnitude was confirmed by the exact Friedman permutation test within each dataset: $Q = 16.4$, exact $p = 8.04 \times 10^{-6}$ for ImageNet-100 and $Q = 15.0$, exact $p = 9.47 \times 10^{-5}$ for CIFAR-100. For reference, the corresponding three-degree-of-freedom asymptotic χ^2 approximations yielded $p = 9.39 \times 10^{-4}$ and $p = 1.82 \times 10^{-3}$, respectively. Both exact results remain significant after correction for the two dataset-level omnibus tests. These omnibus tests address heterogeneity of gap magnitude; the uniformly positive direction is described separately by the 48/48 seed-level gaps and the paired effect sizes in Table 2.

These conclusions do not depend on the adversarial lower-envelope construction. In the supplementary no-envelope sensitivity analysis, all 48 gaps remain positive, all within-seed family rank configurations and positive-seed counts are preserved, and the exact Friedman results are unchanged.

5.4. Endpoint sensitivity

Extending the integration endpoint from $\rho_{\text{max}} = 0.09$ to $\rho_{\text{max}} = 0.095$ did not alter the qualitative conclusions. All eight dataset–family combinations retained positive paired gaps for all six seeds. The corresponding unadjusted descriptive two-sided sign-flip result remained $p = 0.03125$ in every case. The noise, adversarial, and gap rankings of the four PE families were unchanged on both datasets. At the extended endpoint, the mean gap ranged from 0.0048

for ImageNet-100 Learned PE to 0.1668 for ImageNet-100 Sinusoidal PE, and from 0.0100 for CIFAR-100 RoPE to 0.0448 for CIFAR-100 Sinusoidal PE. Because the within-seed family rankings were preserved, the exact Friedman permutation results reported in the preceding subsection were unchanged; the asymptotic reference results were unchanged as well. Full endpoint-specific seed-level and aggregate estimates are retained in the reproducibility package.

5.5. Architectural class and random-noise stability are insufficient predictors

The observed family ordering is not explained by a coarse classification of positional encodings according to where they enter the computation graph. Learned and Sinusoidal encodings both act through additive token-space representations, yet their adversarial robustness gaps differ substantially on ImageNet-100. Likewise, RoPE and ALiBi both modify the attention computation rather than the input token embeddings, but they do not exhibit a common adversarial robustness profile.

The broad distinction between embedding-space and attention-space positional mechanisms is therefore insufficient to determine empirical adversarial positional robustness within the examined cohort. Random-noise stability is also insufficient: the family-specific noise ordering does not consistently preserve the adversarial ordering, and the extreme ranks of the two orderings reverse over the wider CIFAR-100 displacement interval.

These failures indicate that vulnerability cannot be reduced to the location of the positional operation or to average stability under isotropic perturbations. A more informative analysis must separate how efficiently particular parameter-space directions produce attention-logit displacement from how harmful the resulting displacement is for the task. The local directional probe in the following section examines these two factors separately.

5.6. Wider-range rank reversal on CIFAR-100

The difference between the two perturbation regimes becomes especially clear when the CIFAR-100 analysis is extended beyond the primary small-displacement interval to $\rho_{\text{rel}} \in [0, 0.23]$, the conservative common-support endpoint defined in Section 4.8. Table 3 reports the corresponding seed-aggregated nAUC values computed from the final merged measurement grid without extrapolation.

The extreme ranks reverse consistently across all six seeds: Learned PE attains a higher wider-range nAUC than ALiBi under random noise in six of six seeds, whereas ALiBi attains a higher nAUC than Learned PE under

Table 3: Wider-range CIFAR-100 ordering-sensitivity analysis over $\rho_{\text{rel}} \in [0, 0.23]$, the conservative common-support endpoint defined in Section 4.8. Values are mean $\pm$ sample SD of nAUC across the six training seeds. Random-noise curves use the same budget-paired ten-draw aggregation as the primary analysis. Both regimes are generated by the locked canonical analysis pipeline from the final merged measurement grid, including the completed six-seed transition points, without extrapolation. Per-seed values and input SHA-256 hashes are provided in the reproducibility package.

PE family	Random-noise nAUC	Adversarial nAUC
Learned	0.9976 ± 0.0025	0.7106 ± 0.0523
Sinusoidal	0.9922 ± 0.0008	0.7581 ± 0.0390
RoPE	0.9948 ± 0.0009	0.7912 ± 0.1205
ALiBi	0.9641 ± 0.0062	0.8475 ± 0.0146

adversarial perturbations in six of six seeds. At the family-mean level, Learned PE has the highest random-noise nAUC and the lowest adversarial nAUC, while ALiBi shows the opposite mean ordering.

The ordering of the two intermediate families is less stable: RoPE exceeds Sinusoidal in mean adversarial nAUC, but in only two of six individual seeds. RoPE also exhibits by far the largest adversarial seed variability in the cohort (sample SD 0.1205). We therefore do not interpret the complete four-family adversarial ranking as a stable result; the reversal claim concerns the extremes.

This extremal rank reversal provides a stronger form of evidence than a difference in robustness magnitude alone: the choice of perturbation regime changes the qualitative conclusion about which PE family is the most, and which the least, robust at the family-mean level. Consistent with the treatment of the family-specific sign-flip tests in Section 4.8, the 6/6 pairwise reversal counts are reported descriptively rather than as multiplicity-adjusted tests.

6. Factorial Architecture–Precision Decomposition and Objective-Specific Audits

The three cross-cohort contrasts answer different questions and are interpreted jointly rather than as independent robustness tests. Table 4 makes explicit which factor is held fixed in each comparison.

6.1. Canonical four-family cross-cohort comparison

This comparison uses the canonical ViT-B/16 AMP-trained and ViT-S/16 full-FP32-trained cohorts. It is informative about transfer between the two

Table 4: Factorial interpretation of the three ImageNet-100 cross-cohort comparisons. Equal seed labels are alignment labels, not strict training pairs.

Analysis	Architecture contrast	Training contrast	Shared PE families	Inferential role
Canonical	ViT-B vs ViT-S	AMP vs full FP32	4	System-level replication; factors confounded
Within-ViT-S regime	ViT-S fixed	AMP vs full FP32	3	Within-backbone training-regime comparison
AMP-matched architecture	ViT-B vs ViT-S	AMP fixed	3	Cross-architecture effect with training regime fixed

Table 5: Cross-architecture ImageNet-100 nAUC over the locked common-support interval $\rho_{\text{rel}} \in [0, 0.07924]$. Values are mean $\pm$ sample SD across six seeds.

Architecture	PE family	Noise nAUC	Adversarial nAUC	Gap
ViT-B/16	Learned	0.9983 ± 0.0010	0.9945 ± 0.0023	0.0039 ± 0.0024
	Sinusoidal	0.9984 ± 0.0010	0.9034 ± 0.0105	0.0950 ± 0.0110
	RoPE	0.9992 ± 0.0005	0.9669 ± 0.0083	0.0323 ± 0.0086
	ALiBi	0.9685 ± 0.0061	0.8701 ± 0.0185	0.0983 ± 0.0201
ViT-S/16	Learned	0.9981 ± 0.0007	0.8509 ± 0.0374	0.1471 ± 0.0374
	Sinusoidal	0.9988 ± 0.0011	0.7664 ± 0.0385	0.2324 ± 0.0390
	RoPE	0.9992 ± 0.0008	0.8662 ± 0.0397	0.1330 ± 0.0396
	ALiBi	0.9717 ± 0.0070	0.9046 ± 0.0064	0.0670 ± 0.0066

complete canonical systems, but architecture and training precision jointly vary and are not assigned separate causal interpretations. The ImageNet-100 comparison is integrated only over the measured common-support interval $\rho_{\text{rel}} \in [0, 0.0792383973]$. This endpoint is set by the smallest canonical support among all 96 architecture–family–seed–regime curves. Points from the objective-specific all-restart and direct-displacement audits are excluded from this domain.

The random-noise ordering is identical at both scales: RoPE > Sinusoidal > Learned > ALiBi. The adversarial ordering, however, changes from Learned > RoPE > Sinusoidal > ALiBi for ViT-B/16 to ALiBi > RoPE > Learned > Sinusoidal for ViT-S/16. Thus, architecture scale leaves the random ranking almost unchanged while altering which positional mechanism is most stable under task-aligned perturbations.

For all eight combinations of architecture and family, the noise-minus-adversarial gap is positive in all six seeds. Exact Friedman permutation tests show between-family heterogeneity of gap magnitude for ViT-B/16 ($Q = 17.0$, $p = 2.386 \times 10^{-6}$) and ViT-S/16 ($Q = 15.0$, $p = 9.469 \times 10^{-5}$). The same family ordering and omnibus statistics are retained at 0.9 and 0.75 of the common endpoint. The cross-architecture mean differences are reported as

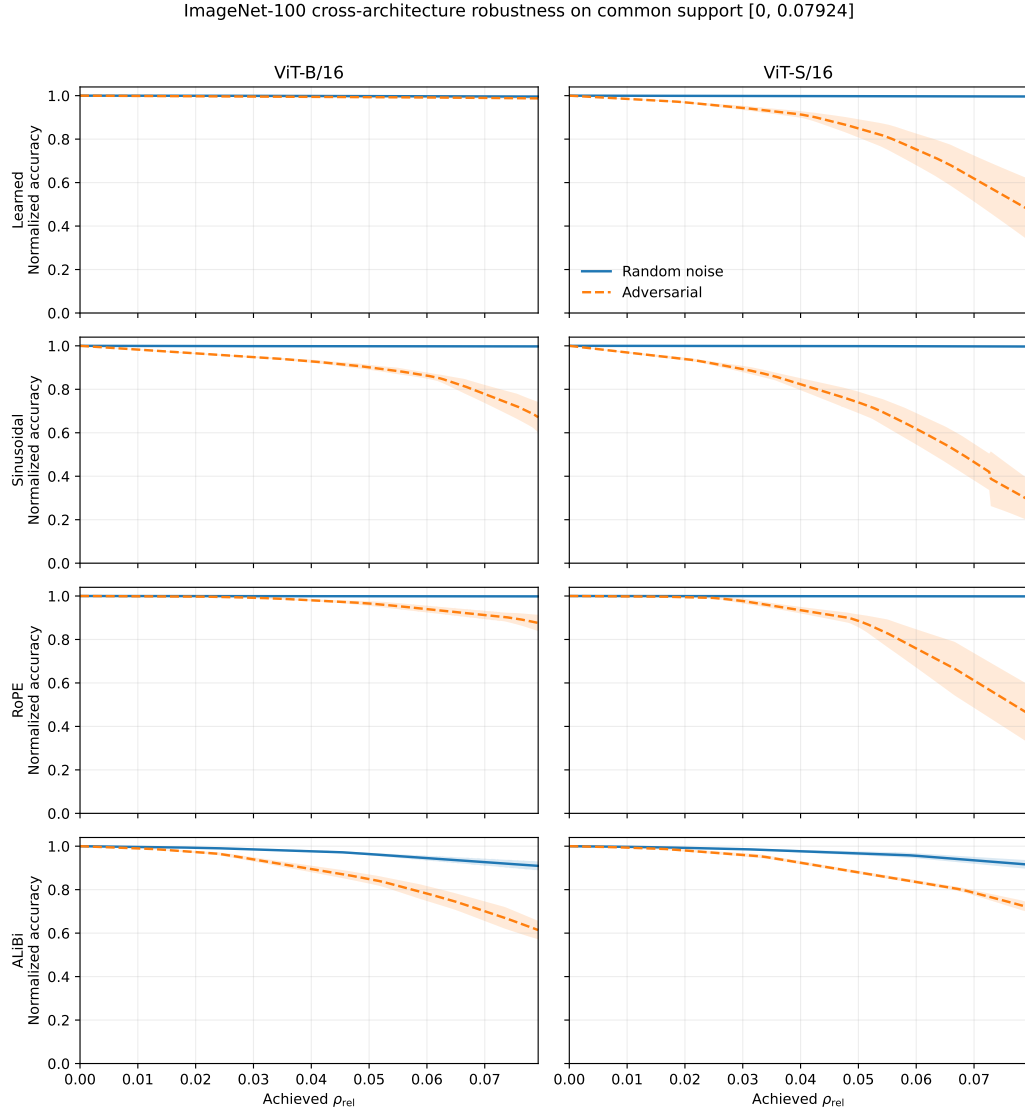


Figure 2: ImageNet-100 canonical robustness curves over the architecture-common support. Lines show across-seed means and shaded regions show ± 1 sample SD. Random noise is solid and task-loss adversarial perturbation is dashed. Every curve is constructed at seed level before aggregation; no point from the objective-specific all-restart or direct-displacement audits is included.

seed-aligned descriptive contrasts, not strict paired effects.

6.2. Within-ViT-S training-regime comparison

We compared the completed ViT-S/16 cohorts trained with CUDA AMP-FP16 and full-FP32 using the same six seed labels and the same held-out split, attack budgets, parameter norm, restart count, and family-specific PGD schedules for Learned, Sinusoidal, and RoPE. Because bit-identical initialization hashes, training-sampler hashes, and per-run GradScaler skip counts were not packaged, we describe this as a *seed-matched*, rather than strictly paired, comparison.

The primary endpoint was $\rho(\varepsilon_{50})$, obtained by interpolating ρ through the same ε bracket in which the monotone accuracy envelope crossed 50% of clean accuracy. Before calculating cohort differences, we fixed an absolute equivalence margin of ± 0.01 in ρ_{rel} ; equivalence required the two-sided 90% seed-matched confidence interval for the mean AMP-minus-FP32 difference to lie fully inside this margin.

Sinusoidal satisfied this criterion (AMP 0.06745 ± 0.00246 versus FP32 0.06797 ± 0.00508 ; $\Delta = -0.00052$, 90% CI $[-0.00498, 0.00394]$). Learned ($\Delta = 0.03447$, 90% CI $[-0.02455, 0.09350]$) and RoPE ($\Delta = 0.00647$, 90% CI $[-0.00448, 0.01742]$) were inconclusive. The aggregate ordering was nevertheless identical in both cohorts, Learned $>$ RoPE $>$ Sinusoidal for $\rho(\varepsilon_{50})$. Support-matched attack and random-noise nAUC rankings were also preserved at $\rho_{\text{max}} = 0.0792384$ and at the 0.9 and 0.75 sensitivity endpoints. Thus the data support descriptive ranking stability and a family-specific Sinusoidal equivalence result, but not a general claim that robustness transfers unchanged between AMP and full-FP32 training.

6.3. AMP-matched cross-architecture comparison

The within-ViT-S/16 comparison shows what changes when the training regime varies at fixed architecture. The AMP-matched comparison instead holds the training regime fixed and varies the backbone. The symmetric comparison contains Learned, Sinusoidal, and RoPE for the same six seed labels; ALiBi is excluded because ViT-S/16 AMP/BF16 training did not yield a valid checkpoint cohort. Attacks and evaluation are FP32 for both architectures. The support intersection over both architectures, all three families, all six seeds, and both perturbation regimes is $\rho_{\text{cross}} = 0.0873456$.

With six nonzero seed contrasts, attainable two-sided exact sign-flip p -values are quantized in steps of $2/64 = 0.03125$. Thus 0.03125 is the floor

Table 6: AMP-matched ImageNet-100 adversarial nAUC over $\rho_{\text{rel}} \in [0, 0.08735]$. Architecture values are mean $\pm$ sample SD across six seeds; Δ is ViT-S minus ViT-B. Exact sign-flip values are secondary directional-consistency checks.

PE family	ViT-B adversarial nAUC	ViT-S adversarial nAUC	Δ adversarial	Seed-aligned 95% CI	Seeds $\Delta < 0$	Exact p	Δ noise nAUC
Learned	0.9937 $\pm$ 0.0024	0.7751 $\pm$ 0.0717	-0.2186	[-0.2929, -0.1443]	6/6	0.03125	-0.000894
Sinusoidal	0.8736 $\pm$ 0.0181	0.7137 $\pm$ 0.0198	-0.1599	[-0.1842, -0.1356]	6/6	0.03125	+0.000729
RoPE	0.9559 $\pm$ 0.0119	0.8627 $\pm$ 0.0557	-0.0932	[-0.1604, -0.0261]	5/6	0.06250	+0.000132

and 0.0625 the adjacent rung, not a finely resolved distinction between the families. Per-family p -values are therefore not treated as primary evidence; effect sizes, intervals, seed signs, and the omnibus interaction carry the interpretation. Unpaired Welch sensitivity intervals are nearly identical to the seed-aligned intervals and remain below zero for all three families (Supplementary Table S1).

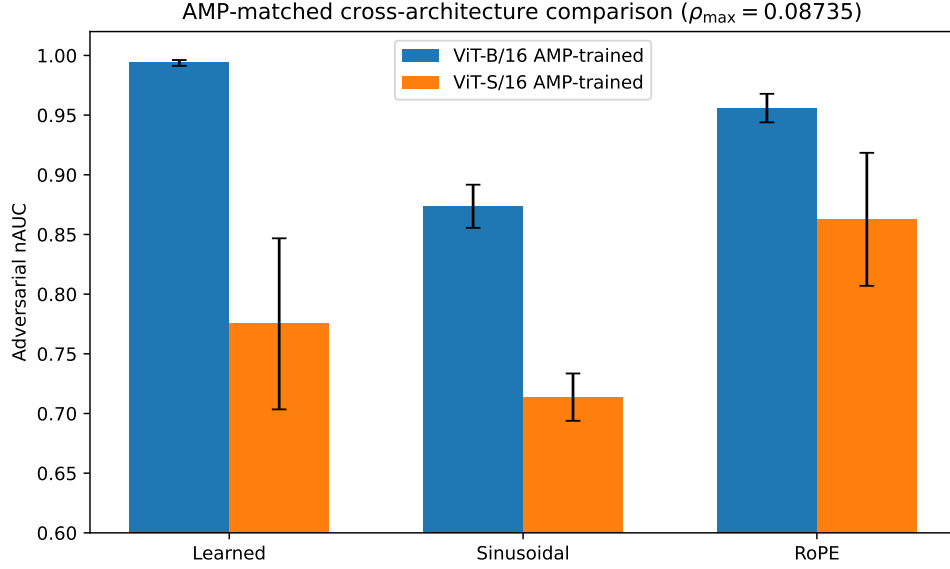


Figure 3: AMP-matched cross-architecture adversarial nAUC over $\rho_{\text{rel}} \in [0, 0.08735]$. Bars and error bars show the across-seed mean and sample standard deviation.

The adversarial ranking changes from Learned $>$ RoPE $>$ Sinusoidal in ViT-B/16 to RoPE $>$ Learned $>$ Sinusoidal in ViT-S/16. Letting Δ denote ViT-S minus ViT-B adversarial nAUC, 5/6 seeds share the ordinal pattern Learned $<$ Sinusoidal $<$ RoPE; seed 42 is the sole exception. The corresponding rank sums are 8, 11, and 17, and the exact Friedman test gives

$Q = 7.0$, $p = 0.02894$.

The interaction is also sign-consistent across pairwise architecture-effect contrasts. Learned minus RoPE has 95% CI $[-0.2353, -0.0155]$ with the same negative sign in 5/6 seeds; Sinusoidal minus RoPE has CI $[-0.1333, -0.00014]$ and is negative in 6/6; and Learned minus Sinusoidal has CI $[-0.1416, 0.0242]$ and is negative in 5/6. The architecture effect is therefore ordinally robust across all three contrasts rather than resting on one pair.

Backbone scale also changes seed dispersion in a PE-dependent way. The ViT-S/ViT-B adversarial-nAUC SD ratios are 29.41 for Learned, 4.66 for RoPE, and 1.10 for Sinusoidal. For Learned, omitting any one seed leaves the SD ratio between 19.93 and 47.27, so the dispersion contrast is not attributable to one isolated run. Its ordering, Learned $\gg$ RoPE $>$ Sinusoidal, differs from the mean-effect ordering, Learned $>$ Sinusoidal $>$ RoPE. We report this as a descriptive second PE-dependent architecture effect without assigning a mechanism.

The random-noise control has no common seed-level direction: noise nAUC is lower in ViT-S for 3/6 Learned, 2/6 Sinusoidal, and 3/6 RoPE seeds, compared with 6/6, 6/6, and 5/6 negative adversarial contrasts. One ViT-S Sinusoidal noise nAUC is 1.00094 because normalized accuracy is not clipped at one and the noisy model marginally exceeds its clean reference on the finite evaluation split.

Threshold results remain consistent with the nAUC analysis. For RoPE, mean ϵ_{50} is essentially unchanged (0.067108 versus 0.067075), but mean ρ_{50} decreases from 0.116608 in ViT-B/16 to 0.083618 in ViT-S/16. For Sinusoidal, both ϵ_{50} and ρ_{50} decrease. The ViT-B/16 Learned curve does not bracket 50% normalized accuracy in three seeds and does not bracket 20% in any seed; consequently, a full $n = 6$ Learned threshold contrast is not reported. This censoring does not affect the measured-support nAUC analysis.

These results cover two backbone sizes and seed-aligned, not bit-identically paired, training runs. They establish that the mapping from achieved attention-logit displacement to task damage differs by PE family and backbone in the tested cohorts, not a universal scaling law or a strict causal estimate of model size.

6.4. *The Sinusoidal task-loss frontier is objective-specific*

The all-restart audit records all five Sinusoidal task-loss restarts at 48 combinations of architecture, seed, and budget, completing all 240 planned restarts. Under the task-loss objective, larger native budget can increase loss

Table 7: Same-seed, same-budget comparison for ViT-S/16 FP32 Sinusoidal seed 42.

Budget	Objective	ρ_{rel}	Accuracy (%)
0.0095	direct- ρ	0.10209	76.70
0.0095	task loss	0.09380	7.36
0.0200	direct- ρ	0.29505	19.52
0.0200	task loss	0.07945	1.70

and reduce accuracy while the loss-selected global displacement stagnates or decreases. We therefore call the observed boundary a *Sinusoidal task-loss frontier*; it is not a structural upper bound nor an estimate of the largest attainable displacement.

A separate direct-displacement audit on ViT-S/16 FP32 Sinusoidal seed 42 shows that this frontier is objective-specific. Table 7 uses displacement from the same 256-image calibration split and accuracy from the same fixed 3464-image evaluation split for both objectives. At budget 0.0095, direct-displacement optimization reaches slightly larger global displacement yet preserves 76.70% accuracy, compared with 7.36% under task-loss optimization. At budget 0.020, direct- ρ reaches more than three times the task-loss displacement yet preserves much more accuracy. Global ρ is therefore not a sufficient scalar predictor of functional damage, and its relationship with damage is nonlinear in budget and direction.

The direct- ρ schedule is selected by a prespecified best-of-five criterion. Its best-of-five envelope is stable between 200×5 and 400×5 , but individual restarts do not all converge. The reported ρ is consequently an empirical lower bound on attainable displacement under the tested protocol, not a supremum estimate or evidence that the search was exhausted.

6.5. Layerwise displacement allocation

Direct- ρ perturbations are substantially more concentrated across layers than task-loss perturbations. The effective number of layers is 4.04 versus 9.22 at budget 0.0095 and 4.45 versus 10.98 at budget 0.020. Direct- ρ profiles are dominated by layer 2, whereas the task-loss profiles peak at layer 4; their profile cosines are 0.589 and 0.613. This supports direction-dependent inter-layer redistribution but does not identify particular heads or tokens and does not establish a causal mechanism.

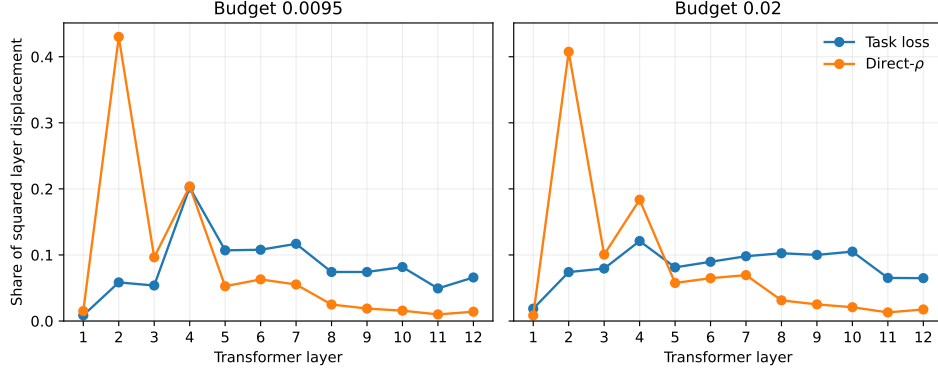


Figure 4: Layerwise share of squared attention-logit displacement for ViT-S/16 FP32 Sinusoidal seed 42. Direct- ρ and task-loss profiles are shown at the same nominal budgets.

7. Local Directional Geometry Probe

7.1. Method

The AUC analysis establishes whether random and adversarial robustness differ, but it does not determine why. We therefore perform a local directional probe around every trained checkpoint.

A task-gradient direction is estimated from a fixed subset $\mathcal{G}$ of 256 attack-split images:

$$g_p = \nabla_p \frac{1}{|\mathcal{G}|} \sum_{(x,y) \in \mathcal{G}} \mathcal{L}(f(x; \theta, p), y). \quad (13)$$

The normalized ascent direction is

$$u_{\text{task}} = \frac{g_p}{\|g_p\|_{\text{RMS}}}. \quad (14)$$

For comparison, we sample 32 Gaussian directions $u_{\text{rand}}^{(j)}$ and normalize each using the same global RMS convention.

For each dataset, the base split is generated once from the deterministic permutation with split seed 0. Denoting the resulting 256-image calibration split by $\mathcal{C}$ and the 1,280-image attack split by $\mathcal{A}$, the task-gradient subset is a fixed 256-image set $\mathcal{G} \subset \mathcal{A}$, whereas the geometry subset is a fixed 64-image set $\mathcal{H} \subset \mathcal{C}$. Consequently, $\mathcal{G} \cap \mathcal{H} = \emptyset$ by construction. The same dataset-specific indices are reused across all PE families and training seeds, preserving paired comparisons.

For each direction, its scalar magnitude is calibrated toward the nominal target

$$\rho_{\text{rel}} = 0.03$$

using the first 32 images of $\mathcal{H}$, denoted $\mathcal{H}_\rho \subset \mathcal{H}$. The calibrated direction is then evaluated on the complete 64-image set $\mathcal{H}$. Accordingly, all reported displacement, cross-entropy change, and damage-efficiency quantities use measurements on $\mathcal{H}$ rather than only on the inner calibration subset $\mathcal{H}_\rho$.

The nominal target is used only to place the directions in a common local neighbourhood. All final normalized quantities use the displacement actually measured over the complete set $\mathcal{H}$.

7.2. Directional functional amplification

For a parameter-space direction u , we define its local functional gain at probe radius r_0 as

$$\gamma(u) = \frac{\rho_{\text{rel}}(r_0 u)}{r_0}. \quad (15)$$

A fixed value $r_0 = 10^{-4}$ is used for every dataset, PE family, training seed, and probe direction. The radius is expressed in the same global-RMS-normalized native parameter-space units used to define the unit direction u and corresponds to the `probe_radius` field in the archived geometry-probe metadata.

The task-to-random amplification ratio is

$$R_\gamma = \frac{\gamma(u_{\text{task}})}{\text{median}_j \gamma(u_{\text{rand}}^{(j)})}. \quad (16)$$

A large value of R_γ indicates that the task-gradient direction changes the attention logits much more efficiently than the median of the 32 sampled Gaussian reference directions at the same parameter-space RMS magnitude.

7.3. Task-selective damage efficiency

Let α_u denote the calibrated scalar for direction u , and let

$$\rho_{\mathcal{H}}(u) = \rho_{\text{rel}}(\alpha_u u; \mathcal{H})$$

be its actually achieved displacement on the complete geometry subset.

The corresponding increase in cross-entropy is

$$\Delta \text{CE}(u) = \text{CE}(p + \alpha_u u; \mathcal{H}) - \text{CE}(p; \mathcal{H}). \quad (17)$$

We define damage efficiency as

$$\eta(u) = \frac{\Delta\text{CE}(u)}{\rho_{\mathcal{H}}(u)}. \quad (18)$$

The task-selective damage gap is

$$\Delta\eta = \eta(u_{\text{task}}) - \text{median}_j \eta(u_{\text{rand}}^{(j)}). \quad (19)$$

Because Eq. (18) uses the actually measured full-subset displacement, it does not assume that every calibrated direction achieves exactly the nominal target value of 0.03.

7.4. Scope of the local probe

The probe is a local directional diagnostic. Here, “held out” refers specifically to separation from the images used to estimate the task-gradient direction. The geometry subset is drawn from the fixed calibration partition and contains the inner magnitude-calibration subset $\mathcal{H}_\rho$; it should not be interpreted as an additional independent test set. The probe does not compute the singular-value decomposition of the full model Jacobian and should not be interpreted as recovering globally dominant sensitivity directions.

A matrix-free spectral analysis using Jacobian–vector and vector–Jacobian products could estimate a small number of leading singular components without explicitly constructing the full Jacobian. Such an analysis would primarily characterise parameter-to-attention amplification. It would not, by itself, establish whether the resulting attention displacement is aligned with task-critical directions.

The present analysis instead directly compares task-aligned and random parameter directions and separately measures their functional amplification and classification damage.

7.5. Directional amplification results

The task-gradient direction exhibits strongly family-dependent functional amplification. Values below are reported as mean $\pm$ sample standard deviation across the six independently trained checkpoints.

On ImageNet-100, the task-to-random functional-gain ratio is $11.40 \pm 3.49\times$ for Learned PE, compared with $1.54 \pm 0.08\times$ for Sinusoidal PE, $1.25 \pm 0.03\times$ for RoPE, and $1.06 \pm 0.03\times$ for ALiBi. On CIFAR-100, the corresponding ratios are $5.81 \pm 0.33\times$, $3.42 \pm 0.14\times$, $1.69 \pm 0.42\times$, and $1.05 \pm 0.09\times$, respectively.

These results show that the local mapping from PE parameter space to attention-logit space is highly anisotropic for Learned PE and, to a lesser extent, CIFAR-100 Sinusoidal PE. In contrast, ALiBi exhibits task and random functional gains of similar magnitude. Functional gain, however, quantifies only how efficiently a direction reaches a displaced attention state; it does not quantify how harmful that state is for classification.

7.6. Task-selective damage results

Geometric amplification and task-selective damage are distinct. The mean task-minus-random damage-efficiency gap is therefore reported together with its sample standard deviation and the number of seeds in which the gap is positive. A positive family mean is not treated as seed-consistent evidence unless the seed signs support that interpretation.

On ImageNet-100, Learned PE combines the strongest amplification, $11.40 \pm 3.49\times$, with a much smaller damage-efficiency gap of 0.91 ± 1.10 , positive in 6/6 seeds. Sinusoidal PE shows the complementary pattern: its amplification is only $1.54 \pm 0.08\times$, yet its damage-efficiency gap is the largest, 6.39 ± 2.91 , and is positive in 6/6 seeds. RoPE has a near-zero mean gap of 0.09 ± 0.33 , positive in only 4/6 seeds; this does not provide stable family-specific evidence of additional task-selective harmfulness. ALiBi has a positive mean gap of 2.25 ± 3.46 , positive in 5/6 seeds, but the large between-seed variation requires a heterogeneous rather than uniform interpretation.

The same separation is especially clear on CIFAR-100. Learned PE has a large task-to-random gain of $5.81 \pm 0.33\times$, whereas its damage-efficiency gap is only 0.14 ± 0.33 and positive in 3/6 seeds. It therefore provides a direct counterexample to the idea that strong parameter-to-attention amplification necessarily entails additional task-selective damage. Sinusoidal, RoPE, and ALiBi show smaller amplification factors but more consistently positive gaps: 0.48 ± 0.13 in 6/6 seeds, 0.47 ± 0.59 in 5/6 seeds, and 0.63 ± 0.50 in 5/6 seeds, respectively.

The local geometry probe therefore distinguishes two measurable components:

1. how efficiently a parameter direction produces an internal attention-logit displacement;
2. how damaging that internal displacement is for the classification task.

Neither quantity alone is sufficient to explain positional-parameter robustness.

7.7. Summary of the geometry analysis

The local probe shows that directional functional amplification and task-selective harmfulness are empirically distinct properties. A PE family can strongly amplify a task-gradient direction in attention-logit space without exhibiting the largest damage per unit of achieved displacement, as observed for Learned PE. Conversely, a family can show only moderate task-to-random amplification while producing highly task-selective internal displacements, as observed for ImageNet-100 Sinusoidal PE.

The geometry results provide a mechanism-oriented interpretation of the robustness curves rather than an independent robustness ranking. They are consistent with family-dependent, anisotropic local mappings from positional parameter space to attention-logit space, together with family- and dataset-dependent alignment between the induced displacement and task-critical directions. Because the probe compares one task-gradient direction with a finite random reference set, these findings should be interpreted as local directional diagnostics rather than as a complete spectral characterization of the Jacobian.

8. Discussion

8.1. Random robustness is not an adversarial-robustness proxy

Across both ViT scales, random-noise nAUC exceeds adversarial nAUC in every family and seed. The result is not merely that optimized attacks are stronger. Gap magnitude and family ordering vary with positional mechanism, dataset, and architecture, so stability under typical Gaussian directions does not identify the family most stable under task-aligned directions.

The canonical cross-cohort replication sharpens this conclusion. Random ordering is preserved from the AMP-trained ViT-B/16 cohort to the full-FP32 ViT-S/16 cohort, while adversarial ordering changes substantially. The Learned–ALiBi extremes reverse: Learned is strongest and ALiBi weakest adversarially for ViT-B/16, but ALiBi is strongest and Learned falls below RoPE for ViT-S/16. This is a descriptive cross-cohort effect in which architecture and training precision jointly vary, not an isolated architecture effect or a universal scaling law.

The three contrasts form a factorial decomposition. The canonical comparison shows system-level transfer when architecture and precision jointly vary; the within-ViT-S/16 comparison holds architecture fixed and varies training regime; and the AMP-matched comparison holds AMP training fixed and

varies architecture. In the AMP-matched comparison, ViT-S/16 adversarial nAUC is lower for all three shared families while random-noise changes have no common seed-level direction. The architecture effect is ordinaly consistent across pairwise family contrasts, reverses the Learned–RoPE ordering, and changes seed dispersion most strongly for Learned. Thus PE family governs both the mean and the variability of how backbone scale maps functional displacement into task damage.

8.2. *Equal displacement magnitude does not imply equal damage*

The achieved ρ_{rel} controls aggregate RMS magnitude of attention-logit change, not orientation. Perturbations with equal global displacement can differ in allocation across layers and heads, query–key structure, token pattern, and alignment with task-critical directions.

The local geometry probe already separates functional amplification from task-selective harmfulness. ImageNet-100 Sinusoidal PE has only $1.54 \pm 0.08 \times$ task-to-random functional gain but the largest damage-efficiency gap, 6.39 ± 2.91 , positive in all six ViT-B/16 seeds. CIFAR-100 Learned PE provides the converse: despite $5.81 \pm 0.33 \times$ amplification, its damage-efficiency gap is only 0.14 ± 0.33 and positive in 3/6 seeds.

The direct-displacement audit provides a stronger objective-level demonstration. For ViT-S/16 Sinusoidal seed 42, direct- ρ optimization at budget 0.0095 achieves slightly larger displacement yet preserves 76.70% accuracy, compared with 7.36% under task-loss optimization. At budget 0.020, direct- ρ reaches much larger displacement but remains less destructive. The layerwise profile analysis shows that these directions also distribute displacement differently across blocks. Magnitude, orientation, and task damage are therefore separate quantities.

8.3. *Family-specific and architecture-specific geometry*

The four families differ in parameter dimension, sharing, and point of injection. Learned embeddings provide a high-dimensional additive space; Sinusoidal perturbation changes deterministic positional state; RoPE perturbation is tied through phase rotations; and ALiBi uses a low-dimensional slope representation shared across blocks. These structural differences make architecture changes consequential: changing embedding dimension and head count changes both the native perturbation space and the mapping into attention, even when head dimension and depth are held fixed.

The strong early-raster prior of the studied ALiBi-style implementation may contribute to its distinctive behavior, but the experiments do not isolate that mechanism causally. Likewise, the direct- ρ layer profile is descriptive and cannot establish which heads, tokens, or downstream computations cause the observed accuracy changes.

8.4. *Implications for robustness evaluation*

A defensible positional-robustness evaluation should report random and task-aligned perturbations separately, reparameterize both by achieved functional displacement, and restrict integration to measured common support. Architecture comparisons should use the intersection of support for exactly the curves being compared and should not import larger points from a different objective. Objective-specific audits, including the all-restart and direct-displacement analyses, are valuable structural evidence, but must remain separate from the canonical task-loss AUC domain.

Family summaries should include seed-level effects, mean $\pm$ sample SD, sign counts, and dispersion. With six nonzero contrasts, two-sided exact sign-flip p -values are quantized in steps of $2/64$: 0.03125 is the floor and 0.0625 the adjacent rung. The omnibus interaction, effect magnitudes, interval agreement, ordinal consistency, and seed dispersion are therefore more informative than treating family-wise p -values as finely resolved evidence.

8.5. *Limitations*

The canonical study covers two datasets for ViT-B/16 but only ImageNet-100 for ViT-S/16. Two architecture sizes cannot establish a general scaling law. Equal seed labels across architectures do not form strict training pairs because the backbones consume different random streams during initialization and do not guarantee identical data order; unpaired Welch sensitivity intervals nevertheless agree with the seed-aligned intervals. The canonical four-family comparison also confounds architecture with training precision; the AMP-matched comparison removes that confound for three families but cannot include ALiBi because the ViT-S/16 AMP training cohort collapsed. Additional model sizes, patch resolutions, and datasets are required for broader generalization.

All four baseline constructions use one-dimensional row-major patch ordering. The conclusions do not establish the behavior of axial, factorized, or explicitly two-dimensional variants. AdamW weight decay also acts on

the trainable Learned positional table but not deterministic buffers, so comparisons characterize the complete as-trained systems rather than a weight-decay-invariant causal effect of positional form.

Canonical adversarial perturbations are optimized under native family-specific parameter constraints and reparameterized afterward by achieved ρ_{rel} ; they are not directly ρ -constrained. The direct-displacement audit partially probes this distinction but is restricted to ViT-S/16 FP32 Sinusoidal seed 42. Its schedule-level PASS concerns the best-of-five envelope, not convergence of every restart or exhaustion of the search. No cross-seed or cross-architecture direct- ρ generalization is made.

The scalar displacement aggregates layers, heads, queries, and keys. The layerwise displacement-profile analysis resolves the layer dimension for one objective comparison but does not resolve heads or tokens and is not causal. The local geometry probe uses one task-gradient direction and 32 Gaussian directions rather than a complete Jacobian spectrum. With six checkpoints per family, mixed-sign family-specific damage summaries remain descriptive.

9. Conclusion

We compared Learned, Sinusoidal, RoPE, and ALiBi under random parameter noise and task-loss adversarial perturbations on a common achieved attention-logit displacement scale. The canonical evidence spans 72 checkpoints: AMP-trained ViT-B/16 on CIFAR-100 and ImageNet-100 and a full-FP32 ViT-S/16 ImageNet-100 replication, with six seeds per family and cohort. A separate 18-checkpoint ViT-S/16 AMP cohort enables within-backbone precision bridging and AMP-matched cross-architecture analysis.

For ViT-B/16, every dataset–family–seed comparison has a positive noise-minus-adversarial nAUC gap over the primary interval $[0, 0.09]$. Over the architecture-common ImageNet-100 support $[0, 0.07924]$, the gap remains positive in all six seeds for every family at both model scales, with significant between-family heterogeneity. The random-noise ranking is stable across the canonical cohorts, but adversarial ordering is not: Learned is strongest for ViT-B/16, whereas ALiBi is strongest for the full-FP32 ViT-S/16 cohort and Sinusoidal becomes weakest. In the AMP-matched three-family comparison, all ViT-S/16 adversarial nAUC means are lower than their ViT-B/16 counterparts, random nAUC changes have no common seed-level direction, and Learned and RoPE reverse order. The architecture effect is ordinally

consistent in 5/6 seeds and is accompanied by strongly PE-dependent seed dispersion.

The local geometry probe, all-restart audit, direct-displacement audit, and layerwise displacement profiles jointly show why a single scalar robustness label is inadequate. Parameter-to-attention amplification, global displacement, inter-layer allocation, and classification damage can vary independently. The Sinusoidal boundary observed under task-loss optimization is therefore reported explicitly as a task-loss frontier, and direct- ρ points are kept outside canonical support.

Random-noise stability is not a reliable proxy for empirical adversarial positional-parameter robustness. Positional-encoding families differ in how achieved attention-logit displacement is converted into functional damage, and that mapping changes with backbone architecture. Evaluation should therefore combine random and task-aligned directions, achieved functional displacement, measured common-support integration, seed-level dispersion, and explicit objective and architecture qualifiers.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The accompanying reproducibility package contains the manuscript source, analysis code, immutable input hashes, seed-level and aggregate numerical outputs, figures, protocol and split manifests, and the all-restart, direct-displacement, and layerwise audit records used in this article. Trained checkpoint binaries are not duplicated in the manuscript archive; their identities and SHA-256 hashes are recorded. The package will be made publicly available upon publication.

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